

## ONGUDAY, Russian Federation

WMO: 362310

Lat: 50.746N

Lon: 86.142E

Elev: 833

StdP: 91.71

Time Zone: 7.00 (E07)

Period: 82-94

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -33.3      | -30.8 | -35.5                         | 0.1 | -33.0 | -33.1 | 0.2 | -30.2 | 5.1                   | -9.8 | 3.6 | -7.1 | 0.0                   | N/A  | 0.589 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.1                   | 27.8            | 16.8 | 25.8 | 16.1 | 24.0 | 15.4 | 18.5                | 24.2 | 17.4 | 23.6 | 16.4 | 22.4 | 0.9                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.7                            | 13.2 | 20.5 | 15.2 | 11.9 | 19.1 | 14.0 | 11.0 | 18.6 | 56.2          | 24.2 | 52.2 | 23.9 | 49.2 | 22.4 | 25.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 5.5           | 3.9  | 2.9 | -34.6                      | 33.0 | 3.1                | 4.6 | -36.8                                              | 36.4 | -38.6      | 39.1 | -40.3      | 41.7 | -42.5      | 45.0 |    |    |
| (5)               |      |     | -34.8                      | 21.7 | 3.2                | 2.6 | -37.2                                              | 23.6 | -39.0      | 25.1 | -40.8      | 26.6 | -43.1      | 28.5 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 0.7    | -17.7 | -15.6 | -5.9  | 3.7   | 10.7  | 14.7  | 17.1  | 14.9  | 9.3   | 2.1   | -10.1 | -16.0 |  |  |
|                                                                        | DBStd       | 13.64  | 6.02  | 6.24  | 6.36  | 4.32  | 3.63  | 3.41  | 2.68  | 2.50  | 3.23  | 5.53  | 8.75  | 7.16  |  |  |
|                                                                        | HDD10.0     | 4004   | 858   | 717   | 492   | 191   | 37    | 3     | 0     | 1     | 50    | 248   | 602   | 804   |  |  |
|                                                                        | HDD18.3     | 6458   | 1117  | 950   | 751   | 438   | 237   | 115   | 54    | 109   | 270   | 503   | 852   | 1063  |  |  |
|                                                                        | CDD10.0     | 614    | 0     | 0     | 0     | 3     | 59    | 145   | 221   | 153   | 30    | 4     | 0     | 0     |  |  |
|                                                                        | CDD18.3     | 26     | 0     | 0     | 0     | 0     | 7     | 16    | 3     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 531    | 0     | 0     | 0     | 1     | 28    | 147   | 231   | 119   | 5     | 0     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 104    | 0     | 0     | 0     | 0     | 4     | 27    | 53    | 21    | 0     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 0.6    | 0.2   | 0.4   | 0.6   | 1.2   | 1.0   | 0.7   | 0.6   | 0.5   | 0.5   | 0.5   | 0.4   | 0.2   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 363    | 6     | 6     | 6     | 19    | 43    | 67    | 72    | 63    | 38    | 21    | 11    | 10    |  |  |
|                                                                        | PrecMax     | 551    | 18    | 29    | 30    | 61    | 100   | 149   | 147   | 141   | 70    | 78    | 24    | 44    |  |  |
|                                                                        | PrecMin     | 191    | 0     | 0     | 1     | 1     | 11    | 27    | 24    | 14    | 13    | 0     | 0     | 0     |  |  |
|                                                                        | PrecStd     | 73     | 5     | 6     | 5     | 13    | 19    | 24    | 29    | 26    | 16    | 15    | 6     | 9     |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | -1.4  | 1.6   | 14.6  | 21.0  | 26.0  | 29.8  | 31.6  | 29.2  | 24.2  | 19.5  | 10.0  | 4.5   |  |  |
|                                                                        |             | MCWB   | -3.9  | -1.8  | 6.2   | 11.1  | 14.4  | 18.3  | 18.7  | 16.4  | 13.8  | 11.7  | 5.1   | 1.4   |  |  |
|                                                                        | 2%          | DB     | -3.7  | -1.5  | 9.3   | 18.6  | 23.4  | 26.9  | 28.1  | 26.6  | 21.6  | 15.9  | 6.1   | -0.6  |  |  |
|                                                                        |             | MCWB   | -5.6  | -4.4  | 3.5   | 9.5   | 13.1  | 16.9  | 17.0  | 16.0  | 12.4  | 9.8   | 2.2   | -2.7  |  |  |
|                                                                        | 5%          | DB     | -6.3  | -4.0  | 5.9   | 15.8  | 21.5  | 25.0  | 26.0  | 24.4  | 19.4  | 12.8  | 3.1   | -3.5  |  |  |
|                                                                        |             | MCWB   | -7.9  | -6.1  | 1.4   | 8.2   | 12.3  | 15.9  | 16.7  | 15.4  | 11.7  | 7.6   | 0.3   | -4.9  |  |  |
|                                                                        | 10%         | DB     | -8.7  | -6.4  | 3.2   | 13.4  | 19.4  | 22.4  | 23.9  | 22.2  | 17.1  | 9.9   | 0.8   | -5.8  |  |  |
|                                                                        |             | MCWB   | -9.8  | -8.0  | -0.2  | 6.8   | 11.3  | 14.9  | 16.2  | 14.8  | 10.7  | 5.9   | -1.9  | -7.1  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | -3.4  | -1.6  | 6.6   | 12.5  | 15.4  | 20.5  | 21.0  | 18.6  | 16.3  | 12.4  | 5.6   | 0.9   |  |  |
|                                                                        |             | MCDB   | -1.7  | 1.7   | 14.5  | 19.5  | 24.9  | 26.9  | 24.9  | 23.7  | 20.1  | 17.7  | 9.2   | 4.1   |  |  |
|                                                                        | 2%          | WB     | -5.6  | -4.2  | 3.8   | 10.3  | 13.8  | 18.0  | 19.0  | 17.2  | 13.7  | 10.3  | 3.0   | -2.4  |  |  |
|                                                                        |             | MCDB   | -3.8  | -1.7  | 8.8   | 17.4  | 21.9  | 24.4  | 24.4  | 24.3  | 19.2  | 15.2  | 5.8   | -0.9  |  |  |
|                                                                        | 5%          | WB     | -7.8  | -6.2  | 1.7   | 8.5   | 12.8  | 16.7  | 17.7  | 16.2  | 12.2  | 8.1   | 0.4   | -4.9  |  |  |
|                                                                        |             | MCDB   | -6.4  | -4.1  | 5.4   | 14.9  | 20.3  | 23.0  | 23.5  | 22.7  | 17.8  | 12.0  | 2.9   | -3.3  |  |  |
|                                                                        | 10%         | WB     | -9.8  | -8.0  | 0.0   | 6.9   | 11.7  | 15.6  | 16.8  | 15.3  | 11.0  | 6.3   | -1.8  | -7.1  |  |  |
|                                                                        |             | MCDB   | -8.6  | -6.5  | 3.0   | 12.8  | 18.2  | 21.4  | 22.8  | 21.0  | 16.4  | 9.6   | 0.5   | -5.8  |  |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 9.2   | 12.0  | 13.2  | 14.5  | 13.6  | 12.6  | 12.1  | 13.1  | 13.3  | 11.8  | 9.2   | 8.6   |  |  |
|                                                                        | 5% DB       | MCDBR  | 10.2  | 12.6  | 15.6  | 19.9  | 18.7  | 17.4  | 16.7  | 18.8  | 18.7  | 16.6  | 10.7  | 11.4  |  |  |
|                                                                        |             | MCWBR  | 8.7   | 10.2  | 10.3  | 12.1  | 10.2  | 8.6   | 7.5   | 9.2   | 10.6  | 10.8  | 7.8   | 9.1   |  |  |
|                                                                        | 5% WB       | MCDBR  | 10.2  | 12.1  | 14.9  | 17.9  | 17.1  | 15.5  | 13.9  | 15.8  | 16.5  | 15.6  | 10.1  | 10.4  |  |  |
|                                                                        |             | MCWBR  | 8.8   | 10.0  | 10.1  | 11.5  | 10.1  | 7.9   | 7.7   | 8.9   | 10.6  | 10.6  | 7.7   | 8.8   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.258 | 0.288 | 0.341 | 0.394 | 0.385 | 0.386 | 0.398 | 0.377 | 0.337 | 0.324 | 0.287 | 0.259 |  |  |
|                                                                        | taud        |        | 2.179 | 2.132 | 2.062 | 2.125 | 2.250 | 2.338 | 2.321 | 2.352 | 2.417 | 2.343 | 2.232 | 2.147 |  |  |
|                                                                        | Ebn at Noon |        | 766   | 833   | 845   | 841   | 869   | 870   | 852   | 853   | 855   | 788   | 717   | 692   |  |  |
|                                                                        | Edh at Noon |        | 72    | 101   | 132   | 140   | 130   | 120   | 120   | 111   | 93    | 82    | 69    | 64    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.40  | 2.49  | 3.99  | 5.02  | 5.58  | 5.92  | 5.56  | 5.00  | 3.77  | 2.23  | 1.33  | 1.02  |  |  |
|                                                                        | RadStd      |        | 0.11  | 0.14  | 0.25  | 0.31  | 0.47  | 0.53  | 0.31  | 0.35  | 0.33  | 0.25  | 0.12  | 0.10  |  |  |

## Historical Trends

|                             |  | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |         |         |         |
|-----------------------------|--|-------|---------|--------|-------|---------|-------|-----|-------------|---------|---------|---------|
|                             |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)     | (m)     |         |
| (46) Station Only           |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A     | N/A     |         |
| (47) Regional (6 neighbors) |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A     | N/A     |         |

Nomenclature: See separate page